Abhinav Bhatia, Ph.D.

Applied Scientist at Microsoft Xbox Game Studios Redmond
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Last updated: June 26, 2026. Latest CV available at abhinavbhatia.me/#cv

Interests

Reinforcement Learning, Offline RL, Real-time Planning, AI Safety, Autonomous Vehicles, Embodied Robotics, LLM-Assisted Decision Making, Vision-Language-Action models

Education

University of Massachusetts Amherst — 3.95/4
MS/Ph.D. in Computer Science, advised by Prof. Shlomo Zilberstein — 2026

Birla Institute of Technology and Science (BITS) - Pilani, Pilani Campus — 9.27/10
B.E. (Hons.) in Computer Science — 2015

Experience

Applied Scientist — June 2026 – Present
Xbox Game Studios, Microsoft, Redmond, WA, USA

- Developing AI for automated game-playing.

Applied Sciences Intern — June 2024 – August 2024
Xbox Game Studios, Microsoft, Redmond, WA, USA

- Multi-task inverse and offline reinforcement learning for text conditioned automated gameplay.

Research Engineer — June 2017 – July 2019
School of Computing and Information Sciences, Singapore Management University, Singapore

- Supervised by Prof. Pradeep Varakantham and Prof. Akshat Kumar.
- Worked on optimizing constrained resource allocation at city scale with semireal datasets using deep reinforcement learning.

Software Engineer — August 2015 – June 2017
WalmartLabs, Bengaluru, India

- Was part of the *Operations, Analytics & Research* team in the supply-chain division of Walmart eCommerce.
- Developed an Elasticsearch-based distributed database for data analysis.
- Developed a deep-learning-based system for anomaly detection in large, live incoming data streams.

Software Development Engineering Intern — January 2015 – June 2015
Amazon, Bengaluru, India

- Worked on offline experience for Prime Video.
- Worked on optimizing content load time for Prime Video on Kindle tablets.

Relevant Publications

Abhinav Bhatia, Samer B. Nashed and Shlomo Zilberstein. "RL^3: Boosting Meta Reinforcement Learning via RL inside RL^2". In *NeurIPS Workshop on Generalization in Planning (NeurIPS GenPlan 2023)* and *Reinforcement Learning Conference (RLC 2025)*.

Samer B. Nashed, Justin Svegliato, **Abhinav Bhatia**, Stuart Russell and Shlomo Zilberstein (2022). "Selecting the partial state abstractions of MDPs: A metareasoning approach with deep reinforcement learning". In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022)*.

Abhinav Bhatia, Justin Svegliato, Samer B. Nashed and Shlomo Zilberstein. "Tuning the Hyperparameters of Anytime Planning: A Metareasoning Approach with Deep Reinforcement Learning". In *Proceedings of the International Conference on Automated Planning and Scheduling (ICAPS 2022).*

Abhinav Bhatia, Philip S. Thomas, and Shlomo Zilberstein. "Adaptive Rollout Length for Model-Based RL Using Model-Free Deep RL". In *(ArXiv 2022).*

Abhinav Bhatia, Justin Svegliato and Shlomo Zilberstein. "On the Benefits of Randomly Adjusting Anytime Weighted A*". In *Proceedings of the International Symposium on Combinatorial Search (SoCS 2021).*

Abhinav Bhatia, Pradeep Varakantham and Akshat Kumar. "Resource Constrained Deep Reinforcement Learning". In *Proceedings of the International Conference on Automated Planning and Scheduling (ICAPS 2019).*

Ongoing Projects

Safe Runtime Personalization of Reinforcement Learning Policies

- Developed a meta-reasoning framework that enables users to customize personality of customer-facing autonomous agents, for example, driving style in autonomous vehicles, at runtime through multiple preference variables (resulting in millions of combinations vs. fixed presets) while provably guaranteeing safety even for untested configurations. Increases adoption and addresses safety challenges at scale.
- The approach applies to any robotic or AI assistant.
- As a testbench, designed and implemented a full-stack AV simulator from scratch in pure C, enabling point-to-point driving simulation in seconds. Includes a city map, NPC traffic, and a lidar and vision stack with multiple cameras; supports ego control with deep RL through Python bindings or with lightweight deep-RL implemented in C.

Relevant Ph.D. Coursework

Artificial Intelligence, Reinforcement Learning, Robotics, Advanced Robot Dynamics & Control, Machine Learning, Neural Networks, Advanced Algorithms

Misc: Advanced Robot Dynamics & Control coursework project involved training a simulated humanoid developed by a neighboring lab at UMass in Isaac Sim.

Programming Skills

Languages: Experienced in C, Python, Julia. Familiar with C++ (built full game in undergrad)
Frameworks: PyTorch, OpenAI Gym, Isaac Sim, MuJoCo, Tensorflow, CPLEX, Unity3D

Service & Teaching

- Dissertation Writing Fellowship Awardee, UMass Amherst, Spring 2026.
- Organizing committee member, AAAI 2025 GenPlan workshop.
- Program Committee member: IJCAI 2026, 2025, 2024; NeurIPS GenPlan 2023; AAAI 2023.
- Paper reviewer, JMLR, 2023; AIJ 2021.
- Teaching Assistant, CS383 Artificial Intelligence, UMass Amherst (Fall 2022).

Misc.

- Conceptualized, developed and organized AI bot-making competition for a video game (IEEE BITS-Pilani chapter, 2014).
- 1st Prize - PC 3D Gesture Interface using Kinect (BITS-Pilani Tech Festival 2014).
- Offered Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship, Govt. of India (2010).

Webpage: https://abhinavbhatia.me | **GitHub**: https://github.com/bhatiaabhinav
Google Scholar: https://scholar.google.com/citations?user=Y53CNrIAAAAJ&hl